Part I

Fundamentals

Lecture 1. Matrix-Vector Multiplication

You already know the formula for matrix-vector multiplication. Nevertheless, the purpose of this first lecture is to describe a way of interpreting such products that may be less familiar. If $b = Ax$, then b *is a linear combination of the columns of* A.

Familiar Definitions

Let x be an n-dimensional column vector and let A be an $m \times n$ matrix (m rows, n columns). Then the matrix-vector product $b = Ax$ is the m-dimensional column vector defined as follows:

$$b_i = \sum_{j=1}^{n} a_{ij} x_j, \qquad i = 1, \ldots, m. \tag{1.1}$$

Here b_i denotes the ith entry of b, a_{ij} denotes the i, j entry of A (ith row, jth column), and x_j denotes the jth entry of x. For simplicity, we assume in all but a few lectures of this book that quantities such as these belong to $\mathbb{C}$, the field of complex numbers. The space of m-vectors is $\mathbb{C}^m$, and the space of $m \times n$ matrices is $\mathbb{C}^{m \times n}$.

The map $x \mapsto Ax$ is *linear*, which means that, for any $x, y \in \mathbb{C}^n$ and any $\alpha \in \mathbb{C}$,

$$\begin{aligned} A(x + y) &= Ax + Ay, \\ A(\alpha x) &= \alpha Ax. \end{aligned}$$

Conversely, every linear map from $\mathbb{C}^n$ to $\mathbb{C}^m$ can be expressed as multiplication by an $m \times n$ matrix.

A Matrix Times a Vector

Let a_j denote the jth column of A, an m-vector. Then (1.1) can be rewritten

$$b = Ax = \sum_{j=1}^{n} x_j a_j. \tag{1.2}$$

This equation can be displayed schematically as follows:

$$\begin{bmatrix} \\ b \\ \\ \end{bmatrix} = \left[\begin{array}{c|c|c|c} & & & \\ a_1 & a_2 & \cdots & a_n \\ & & & \end{array}\right] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = x_1 \begin{bmatrix} \\ a_1 \\ \\ \end{bmatrix} + x_2 \begin{bmatrix} \\ a_2 \\ \\ \end{bmatrix} + \cdots + x_n \begin{bmatrix} \\ a_n \\ \\ \end{bmatrix}.$$

In (1.2), b is expressed as a linear combination of the columns a_j. Nothing but a slight change of notation has occurred in going from (1.1) to (1.2). Yet thinking of Ax in terms of the form (1.2) is essential for a proper understanding of the algorithms of numerical linear algebra.

We can summarize these different descriptions of matrix-vector products in the following way. As mathematicians, we are used to viewing the formula $Ax = b$ as a statement that A acts on x to produce b. The formula (1.2), by contrast, suggests the interpretation that x acts on A to produce b.

Example 1.1. Vandermonde Matrix. Fix a sequence of numbers $\{x_1, x_2, \ldots, x_m\}$. If p and q are polynomials of degree $< n$ and α is a scalar, then $p+q$ and αp are also polynomials of degree $< n$. Moreover, the values of these polynomials at the points x_i satisfy the following linearity properties:

$$\begin{aligned} (p+q)(x_i) &= p(x_i) + q(x_i), \\ (\alpha p)(x_i) &= \alpha(p(x_i)). \end{aligned}$$

Thus the map from vectors of coefficients of polynomials p of degree $< n$ to vectors $(p(x_1), p(x_2), \ldots, p(x_m))$ of sampled polynomial values is linear. Any linear map can be expressed as multiplication by a matrix; this is an example. In fact, it is expressed by an $m \times n$ *Vandermonde matrix*

$$A = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_m & x_m^2 & \cdots & x_m^{n-1} \end{bmatrix}.$$

If c is the column vector of coefficients of p,

$$c = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_{n-1} \end{bmatrix}, \qquad p(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_{n-1} x^{n-1},$$

then the product Ac gives the sampled polynomial values. That is, for each i from 1 to m, we have

$$(Ac)_i \;=\; c_0 + c_1 x_i + c_2 x_i^2 + \cdots + c_{n-1} x_i^{n-1} \;=\; p(x_i). \tag{1.3}$$

In this example, it is clear that the matrix-vector product Ac need not be thought of as m distinct scalar summations, each giving a different linear combination of the entries of c, as (1.1) might suggest. Instead, A can be viewed as a matrix of columns, each giving sampled values of a monomial,

$$A = \left[\begin{array}{c|c|c|c|c} 1 & x & x^2 & \cdots & x^{n-1} \end{array}\right], \tag{1.4}$$

and the product Ac should be understood as a single vector summation in the form of (1.2) that at once gives a linear combination of these monomials,

$$Ac \;=\; c_0 + c_1 x + c_2 x^2 + \cdots + c_{n-1} x^{n-1} \;=\; p(x). \qquad \square$$

The remainder of this lecture will review some fundamental concepts in linear algebra from the point of view of (1.2).

A Matrix Times a Matrix

For the matrix-matrix product $B = AC$, *each column of B is a linear combination of the columns of A.* To derive this fact, we begin with the usual formula for matrix products. If A is $\ell \times m$ and C is $m \times n$, then B is $\ell \times n$, with entries defined by

$$b_{ij} = \sum_{k=1}^{m} a_{ik} c_{kj}. \tag{1.5}$$

Here b_{ij}, a_{ik}, and c_{kj} are entries of B, A, and C, respectively. Written in terms of columns, the product is

$$\left[\begin{array}{c|c|c|c} b_1 & b_2 & \cdots & b_n \end{array}\right] = \left[\begin{array}{c|c|c|c} a_1 & a_2 & \cdots & a_m \end{array}\right] \left[\begin{array}{c|c|c|c} c_1 & c_2 & \cdots & c_n \end{array}\right],$$

and (1.5) becomes

$$b_j = Ac_j = \sum_{k=1}^{m} c_{kj} a_k. \tag{1.6}$$

Thus b_j is a linear combination of the columns a_k with coefficients c_{kj}.

Example 1.2. Outer Product. A simple example of a matrix-matrix product is the *outer product*. This is the product of an m-dimensional column vector u with an n-dimensional row vector v; the result is an $m \times n$ matrix of rank 1. The outer product can be written

$$\begin{bmatrix} \\ u \\ \\ \end{bmatrix} \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} = \left[\begin{array}{c|c|c|c} & & & \\ v_1 u & v_2 u & \cdots & v_n u \\ & & & \end{array}\right] = \begin{bmatrix} v_1 u_1 & \cdots & v_n u_1 \\ \vdots & & \vdots \\ v_1 u_m & \cdots & v_n u_m \end{bmatrix}.$$

The columns are all multiples of the same vector u, and similarly, the rows are all multiples of the same vector v. □

Example 1.3. As a second illustration, consider $B = AR$, where R is the upper-triangular $n \times n$ matrix with entries $r_{ij} = 1$ for $i \le j$ and $r_{ij} = 0$ for $i > j$. This product can be written

$$\left[\begin{array}{c|c|c} & & \\ b_1 & \cdots & b_n \\ & & \end{array}\right] = \left[\begin{array}{c|c|c} & & \\ a_1 & \cdots & a_n \\ & & \end{array}\right] \begin{bmatrix} 1 & \cdots & 1 \\ & \ddots & \vdots \\ & & 1 \end{bmatrix}.$$

The column formula (1.6) now gives

$$b_j \;=\; Ar_j \;=\; \sum_{k=1}^{j} a_k. \tag{1.7}$$

That is, the jth column of B is the sum of the first j columns of A. The matrix R is a discrete analogue of an indefinite integral operator. □

Range and Nullspace

The *range* of a matrix A, written range(A), is the set of vectors that can be expressed as Ax for some x. The formula (1.2) leads naturally to the following characterization of range(A).

Theorem 1.1. range(A) *is the space spanned by the columns of* A.

Proof. By (1.2), any Ax is a linear combination of the columns of A. Conversely, any vector y in the space spanned by the columns of A can be written as a linear combination of the columns, $y = \sum_{j=1}^{n} x_j a_j$. Forming a vector x out of the coefficients x_j, we have $y = Ax$, and thus y is in the range of A. □

In view of Theorem 1.1, the range of a matrix A is also called the *column space* of A.

The *nullspace* of $A \in \mathbb{C}^{m\times n}$, written $\text{null}(A)$, is the set of vectors x that satisfy $Ax = 0$, where 0 is the 0-vector in $\mathbb{C}^m$. The entries of each vector $x \in \text{null}(A)$ give the coefficients of an expansion of zero as a linear combination of columns of A: $0 = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n$.

Rank

The *column rank* of a matrix is the dimension of its column space. Similarly, the *row rank* of a matrix is the dimension of the space spanned by its rows. Row rank always equals column rank (among other proofs, this is a corollary of the singular value decomposition, discussed in Lectures 4 and 5), so we refer to this number simply as the *rank* of a matrix.

An $m \times n$ matrix of *full rank* is one that has the maximal possible rank (the lesser of m and n). This means that a matrix of full rank with $m \geq n$ must have n linearly independent columns. Such a matrix can also be characterized by the property that the map it defines is one-to-one.

Theorem 1.2. *A matrix $A \in \mathbb{C}^{m\times n}$ with $m \geq n$ has full rank if and only if it maps no two distinct vectors to the same vector.*

Proof. ($\Longrightarrow$) If A is of full rank, its columns are linearly independent, so they form a basis for $\text{range}(A)$. This means that every $b \in \text{range}(A)$ has a unique linear expansion in terms of the columns of A, and therefore, by (1.2), every $b \in \text{range}(A)$ has a unique x such that $b = Ax$. ($\Longleftarrow$) Conversely, if A is not of full rank, its columns a_j are dependent, and there is a nontrivial linear combination such that $\sum_{j=1}^{n} c_j a_j = 0$. The nonzero vector c formed from the coefficients c_j satisfies $Ac = 0$. But then A maps distinct vectors to the same vector since, for any x, $Ax = A(x + c)$. □

Inverse

A *nonsingular* or *invertible* matrix is a square matrix of full rank. Note that the m columns of a nonsingular $m \times m$ matrix A form a basis for the whole space $\mathbb{C}^m$. Therefore, we can uniquely express any vector as a linear combination of them. In particular, the canonical unit vector with 1 in the jth entry and zeros elsewhere, written e_j, can be expanded:

$$e_j = \sum_{i=1}^{m} z_{ij} a_i. \tag{1.8}$$

Let Z be the matrix with entries z_{ij}, and let z_j denote the jth column of Z. Then (1.8) can be written $e_j = Az_j$. This equation has the form (1.6); it can be written again, most concisely, as

$$\left[\begin{array}{c|c|c} & & \\ e_1 & \cdots & e_m \\ & & \end{array}\right] = I = AZ,$$

where I is the $m \times m$ matrix known as the *identity*. The matrix Z is the *inverse* of A. Any square nonsingular matrix A has a unique inverse, written A^{-1}, that satisfies $AA^{-1} = A^{-1}A = I$.

The following theorem records a number of equivalent conditions that hold when a square matrix is nonsingular. These conditions appear in linear algebra texts, and we shall not give a proof here. Concerning (f), see Lecture 5.

Theorem 1.3. *For $A \in \mathbb{C}^{m\times m}$, the following conditions are equivalent:*
(a) *A has an inverse A^{-1},*
(b) $\text{rank}(A) = m$,
(c) $\text{range}(A) = \mathbb{C}^m$,
(d) $\text{null}(A) = \{0\}$,
(e) *0 is not an eigenvalue of A,*
(f) *0 is not a singular value of A,*
(g) $\det(A) \neq 0$.

Concerning (g), we mention that the determinant, though a convenient notion theoretically, rarely finds a useful role in numerical algorithms.

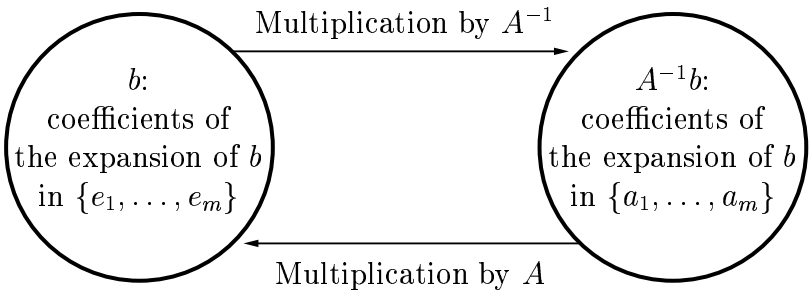
A Matrix Inverse Times a Vector

When writing the product $x = A^{-1}b$, it is important not to let the inverse-matrix notation obscure what is really going on! Rather than thinking of x as the result of applying A^{-1} to b, we should understand it as the unique vector that satisfies the equation $Ax = b$. By (1.2), this means that x is the vector of coefficients of the unique linear expansion of b in the basis of columns of A.

This point cannot be emphasized too much, so we repeat:

$$A^{-1}b \text{ is the vector of coefficients of the expansion of } b \text{ in the basis of columns of } A.$$

Multiplication by A^{-1} is a *change of basis* operation:



In this description we are being casual with terminology, using "b" in one instance to denote an m-tuple of numbers, and in another, as a point in an abstract vector space. The reader should think about these matters until he or she is comfortable with the distinction.

A Note on m and n

Throughout numerical linear algebra, it is customary to take a rectangular matrix to have dimensions $m \times n$. We follow this convention in this book.

What if the matrix is square? The usual convention is to give it dimensions $n \times n$, but in this book we shall generally take the other choice, $m \times m$. Many of our algorithms require us to look at rectangular submatrices formed by taking a subset of the columns of a square matrix. If the submatrix is to be $m \times n$, the original matrix had better be $m \times m$.

Exercises

1.1. Let B be a 4×4 matrix to which we apply the following operations:

1. double column 1,
2. halve row 3,
3. add row 3 to row 1,
4. interchange columns 1 and 4,
5. subtract row 2 from each of the other rows,
6. replace column 4 by column 3,
7. delete column 1 (so that the column dimension is reduced by 1).

(a) Write the result as a product of eight matrices.

(b) Write it again as a product ABC (same B) of three matrices.

1.2. Suppose masses m_1, m_2, m_3, m_4 are located at positions x_1, x_2, x_3, x_4 in a line and connected by springs with spring constants k_{12}, k_{23}, k_{34} whose natural lengths of extension are $\ell_{12}, \ell_{23}, \ell_{34}$. Let f_1, f_2, f_3, f_4 denote the rightward forces on the masses, e.g., $f_1 = k_{12}(x_2 - x_1 - \ell_{12})$.

(a) Write the 4×4 matrix equation relating the column vectors f and x. Let K denote the matrix in this equation.

(b) What are the dimensions of the entries of K in the physics sense (e.g., mass times time, distance divided by mass, etc.)?

(c) What are the dimensions of $\det(K)$, again in the physics sense?

(d) Suppose K is given numerical values based on the units meters, kilograms, and seconds. Now the system is rewritten with a matrix K' based on centimeters, grams, and seconds. What is the relationship of K' to K? What is the relationship of $\det(K')$ to $\det(K)$?

1.3. Generalizing Example 1.3, we say that a square or rectangular matrix R with entries r_{ij} is *upper-triangular* if $r_{ij} = 0$ for $i > j$. By considering what space is spanned by the first n columns of R and using (1.8), show that if R is a nonsingular $m \times m$ upper-triangular matrix, then R^{-1} is also upper-triangular. (The analogous result also holds for lower-triangular matrices.)

1.4. Let $f_1, \ldots, f_8$ be a set of functions defined on the interval $[1, 8]$ with the property that for any numbers $d_1, \ldots, d_8$, there exists a set of coefficients $c_1, \ldots, c_8$ such that

$$\sum_{j=1}^{8} c_j f_j(i) = d_i, \qquad i = 1, \ldots, 8.$$

(a) Show by appealing to the theorems of this lecture that $d_1, \ldots, d_8$ determine $c_1, \ldots, c_8$ uniquely.

(b) Let A be the 8×8 matrix representing the linear mapping from data $d_1, \ldots, d_8$ to coefficients $c_1, \ldots, c_8$. What is the i, j entry of A^{-1}?

Lecture 2. Orthogonal Vectors and Matrices

Since the 1960s, many of the best algorithms of numerical linear algebra have been based in one way or another on orthogonality. In this lecture we present the ingredients: orthogonal vectors and orthogonal (unitary) matrices.

Adjoint

The *complex conjugate* of a scalar z, written $\overline{z}$ or z^*, is obtained by negating its imaginary part. For real z, $\overline{z} = z$.

The *hermitian conjugate* or *adjoint* of an $m \times n$ matrix A, written A^*, is the $n \times m$ matrix whose i,j entry is the complex conjugate of the j,i entry of A. For example,

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \qquad \Longrightarrow \qquad A^* = \begin{bmatrix} \overline{a}_{11} & \overline{a}_{21} & \overline{a}_{31} \\ \overline{a}_{12} & \overline{a}_{22} & \overline{a}_{32} \end{bmatrix}.$$

If $A = A^*$, A is *hermitian*. By definition, a hermitian matrix must be square. For real A, the adjoint simply interchanges the rows and columns of A. In this case, the adjoint is also known as the *transpose*, and is written A^T. If a real matrix is hermitian, that is, $A = A^T$, then it is also said to be *symmetric*.

Most textbooks of numerical linear algebra assume that the matrices under discussion are real and thus principally use ^{T} instead of *. Since most of the ideas to be dealt with are not intrinsically restricted to the reals, however, we have followed the other course. Thus, for example, in this book a row vector

will usually be denoted by, say, a^* rather than a^T. The reader who prefers to imagine that all quantities are real and that * is a synonym for ^{T} will rarely get into trouble.

Inner Product

The *inner product* of two column vectors $x, y \in \mathbb{C}^m$ is the product of the adjoint of x by y:

$$x^*y = \sum_{i=1}^{m} \overline{x}_i y_i. \tag{2.1}$$

The Euclidean length of x may be written $\|x\|$ (vector norms such as this are discussed systematically in the next lecture), and can be defined as the square root of the inner product of x with itself:

$$\|x\| \;=\; \sqrt{x^*x} \;=\; \left(\sum_{i=1}^{m} |x_i|^2 \right)^{1/2}. \tag{2.2}$$

The cosine of the angle α between x and y can also be expressed in terms of the inner product:

$$\cos\alpha = \frac{x^*y}{\|x\|\,\|y\|}. \tag{2.3}$$

At various points of this book, as here, we mention geometric interpretations of algebraic formulas. For these geometric interpretations, the reader should think of the vectors as real rather than complex, although usually the interpretations can be carried over in one way or another to the complex case too.

The inner product is *bilinear*, which means that it is linear in each vector separately:

$$\begin{aligned} (x_1 + x_2)^*y &= x_1^*y + x_2^*y, \\ x^*(y_1 + y_2) &= x^*y_1 + x^*y_2, \\ (\alpha x)^*(\beta y) &= \overline{\alpha}\beta x^*y. \end{aligned}$$

We shall also frequently use the easily proved property that for any matrices or vectors A and B of compatible dimensions,

$$(AB)^* = B^*A^*. \tag{2.4}$$

This is analogous to the equally important formula for products of invertible square matrices,

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{2.5}$$

The notation A^{-*} is a shorthand for $(A^*)^{-1}$ or $(A^{-1})^*$; these two are equal, as can be verified by applying (2.4) with $B = A^{-1}$.

Orthogonal Vectors

A pair of vectors x and y are *orthogonal* if $x^*y = 0$. If x and y are real, this means they lie at right angles to each other in $\mathbb{R}^m$. Two sets of vectors X and Y are orthogonal (also stated "X is orthogonal to Y") if every $x \in X$ is orthogonal to every $y \in Y$.

A set of nonzero vectors S is *orthogonal* if its elements are pairwise orthogonal, i.e., if for $x, y \in S$, $x \neq y \Rightarrow x^*y = 0$. A set of vectors is *orthonormal* if it is orthogonal and, in addition, every $x \in S$ has $\|x\| = 1$.

Theorem 2.1. *The vectors in an orthogonal set S are linearly independent.*

Proof. If the vectors in S are not independent, then some $v_k \in S$ can be expressed as a linear combination of other members $v_1, \ldots, v_n \in S$,

$$v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} c_i v_i.$$

Since $v_k \neq 0$, $v_k^* v_k = \|v_k\|^2 > 0$. Using the bilinearity of inner products and the orthogonality of S, we calculate

$$v_k^* v_k = \sum_{\substack{i=1 \\ i \neq k}}^{n} c_i v_k^* v_i = 0,$$

which contradicts the assumption that the vectors in S are nonzero. □

As a corollary of Theorem 2.1 it follows that if an orthogonal set $S \subseteq \mathbb{C}^m$ contains m vectors, then it is a basis for $\mathbb{C}^m$.

Components of a Vector

The most important idea to draw from the concepts of inner products and orthogonality is this: inner products can be used to decompose arbitrary vectors into orthogonal components.

For example, suppose that $\{q_1, q_2, \ldots, q_n\}$ is an orthonormal set, and let v be an arbitrary vector. The quantity $q_j^* v$ is a scalar. Utilizing these scalars as coordinates in an expansion, we find that the vector

$$r = v - (q_1^* v)q_1 - (q_2^* v)q_2 - \cdots - (q_n^* v)q_n \tag{2.6}$$

is orthogonal to $\{q_1, q_2, \ldots, q_n\}$. This can be verified by computing $q_i^* r$:

$$q_i^* r \;=\; q_i^* v - (q_1^* v)(q_i^* q_1) - \cdots - (q_n^* v)(q_i^* q_n).$$

This sum collapses, since $q_i^* q_j = 0$ for $i \neq j$:

$$q_i^* r \;=\; q_i^* v - (q_i^* v)(q_i^* q_i) \;=\; 0.$$

Thus we see that v can be decomposed into $n+1$ orthogonal components:

$$v \;=\; r + \sum_{i=1}^{n} (q_i^* v) q_i \;=\; r + \sum_{i=1}^{n} (q_i q_i^*) v. \tag{2.7}$$

In this decomposition, r is the part of v orthogonal to the set of vectors $\{q_1, q_2, \ldots, q_n\}$, or, equivalently, to the subspace spanned by this set of vectors, and $(q_i^* v) q_i$ is the part of v in the direction of q_i.

If $\{q_i\}$ is a basis for $\mathbb{C}^m$, then n must be equal to m and r must be the zero vector, so v is completely decomposed into m orthogonal components in the directions of the q_i:

$$v \;=\; \sum_{i=1}^{m} (q_i^* v) q_i \;=\; \sum_{i=1}^{m} (q_i q_i^*) v. \tag{2.8}$$

In both (2.7) and (2.8) we have written the formula in two different ways, once with $(q_i^* v) q_i$ and again with $(q_i q_i^*) v$. These expressions are equal, but they have different interpretations. In the first case, we view v as a sum of coefficients $q_i^* v$ times vectors q_i. In the second, we view v as a sum of orthogonal projections of v onto the various directions q_i. The ith projection operation is achieved by the very special rank-one matrix $q_i q_i^*$. We shall discuss this and other projection processes in Lecture 6.

Unitary Matrices

A square matrix $Q \in \mathbb{C}^{m \times m}$ is *unitary* (in the real case, we also say *orthogonal*) if $Q^* = Q^{-1}$, i.e, if $Q^* Q = I$. In terms of the columns of Q, this product can be written

$$\left[\begin{array}{c} q_1^* \\ \hline q_2^* \\ \hline \vdots \\ \hline q_m^* \end{array}\right] \left[\begin{array}{c|c|c|c} & & & \\ q_1 & q_2 & \cdots & q_m \\ & & & \end{array}\right] \;=\; \left[\begin{array}{cccc} 1 & & & \\ & 1 & & \\ & & \ddots & \\ & & & 1 \end{array}\right].$$

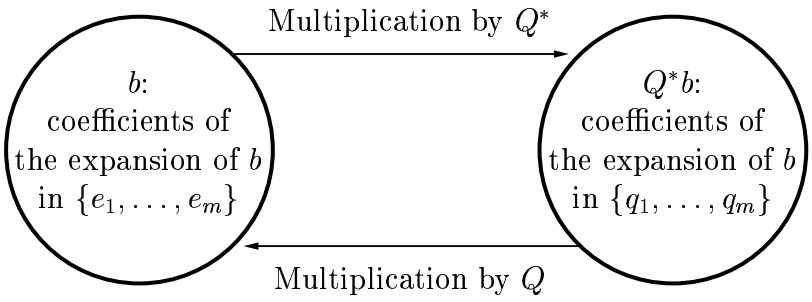
In other words, $q_i^* q_j = \delta_{ij}$, and the columns of a unitary matrix Q form an orthonormal basis of $\mathbb{C}^m$. The symbol δ_{ij} is the *Kronecker delta*, equal to 1 if $i = j$ and 0 if $i \neq j$.

Multiplication by a Unitary Matrix

In the last lecture we discussed the interpretation of matrix-vector products Ax and $A^{-1}b$. If A is a unitary matrix Q, these products become Qx and Q^*b, and the same interpretations are of course still valid. As before, Qx is the linear combination of the columns of Q with coefficients x. Conversely,

Q^*b is the vector of coefficients of the expansion of b
in the basis of columns of Q.

Schematically, the situation looks like this:



These processes of multiplication by a unitary matrix or its adjoint preserve geometric structure in the Euclidean sense, because inner products are preserved. That is, for unitary Q,

$$(Qx)^*(Qy) = x^*y, \tag{2.9}$$

as is readily verified by (2.4). The invariance of inner products means that angles between vectors are preserved, and so are their lengths:

$$\|Qx\| = \|x\|. \tag{2.10}$$

In the real case, multiplication by an orthogonal matrix Q corresponds to a rigid rotation (if $\det Q = 1$) or reflection (if $\det Q = -1$) of the vector space.

Exercises

2.1. Show that if a matrix A is both triangular and unitary, then it is diagonal.

2.2. The Pythagorean theorem asserts that for a set of n orthogonal vectors $\{x_i\}$,

$$\Big\| \sum_{i=1}^{n} x_i \Big\|^2 = \sum_{i=1}^{n} \|x_i\|^2.$$

(a) Prove this in the case $n = 2$ by an explicit computation of $\|x_1 + x_2\|^2$.

(b) Show that this computation also establishes the general case, by induction.

2.3. Let $A \in \mathbb{C}^{m\times m}$ be hermitian. An eigenvector of A is a nonzero vector $x \in \mathbb{C}^m$ such that $Ax = \lambda x$ for some $\lambda \in \mathbb{C}$, the corresponding eigenvalue.

(a) Prove that all eigenvalues of A are real.

(b) Prove that if x and y are eigenvectors corresponding to distinct eigenvalues, then x and y are orthogonal.

2.4. What can be said about the eigenvalues of a unitary matrix?

2.5. Let $S \in \mathbb{C}^{m\times m}$ be *skew-hermitian*, i.e., $S^* = -S$.

(a) Show by using Exercise 2.3 that the eigenvalues of S are pure imaginary.

(b) Show that $I - S$ is nonsingular.

(c) Show that the matrix $Q = (I-S)^{-1}(I+S)$, known as the *Cayley transform* of S, is unitary. (This is a matrix analogue of a linear fractional transformation $(1 + s)/(1 - s)$, which maps the left half of the complex s-plane conformally onto the unit disk.)

2.6. If u and v are m-vectors, the matrix $A = I + uv^*$ is known as a *rank-one perturbation of the identity*. Show that if A is nonsingular, then its inverse has the form $A^{-1} = I + \alpha uv^*$ for some scalar α, and give an expression for α. For what u and v is A singular? If it is singular, what is null(A)?

2.7. A *Hadamard matrix* is a matrix whose entries are all ± 1 and whose transpose is equal to its inverse times a constant factor. It is known that if A is a Hadamard matrix of dimension $m > 2$, then m is a multiple of 4. It is not known, however, whether there is a Hadamard matrix for every such m, though examples have been found for all cases $m \leq 424$.

Show that the following recursive description provides a Hadamard matrix of each dimension $m = 2^k$, $k = 0, 1, 2, \ldots$:

$$H_0 = \begin{bmatrix} 1 \end{bmatrix}, \qquad H_{k+1} = \begin{bmatrix} H_k & H_k \\ H_k & -H_k \end{bmatrix}.$$

Lecture 3. Norms

The essential notions of size and distance in a vector space are captured by norms. These are the yardsticks with which we measure approximations and convergence throughout numerical linear algebra.

Vector Norms

A *norm* is a function $\|\cdot\| : \mathbb{C}^m \to \mathbb{R}$ that assigns a real-valued length to each vector. In order to conform to a reasonable notion of length, a norm must satisfy the following three conditions. For all vectors x and y and for all scalars $\alpha \in \mathbb{C}$,

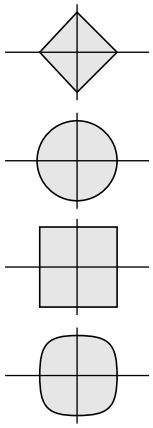
$$\begin{aligned} &(1)\ \ \|x\| \geq 0, \text{ and } \|x\| = 0 \text{ only if } x = 0, \\ &(2)\ \ \|x + y\| \leq \|x\| + \|y\|, \\ &(3)\ \ \|\alpha x\| = |\alpha|\, \|x\|. \end{aligned} \tag{3.1}$$

In words, these conditions require that (1) the norm of a nonzero vector is positive, (2) the norm of a vector sum does not exceed the sum of the norms of its parts—the *triangle inequality*, and (3) scaling a vector scales its norm by the same amount.

In the last lecture, we used $\|\cdot\|$ to denote the Euclidean length function (the square root of the sum of the squares of the entries of a vector). However, the three conditions (3.1) allow for different notions of length, and at times it is useful to have this flexibility.

The most important class of vector norms, the p-norms, are defined below. The closed unit ball $\{x \in \mathbb{C}^m : \|x\| \le 1\}$ corresponding to each norm is illustrated to the right for the case $m = 2$.

$$\begin{aligned}
\|x\|_1 &= \sum_{i=1}^{m} |x_i|, \\
\|x\|_2 &= \left(\sum_{i=1}^{m} |x_i|^2 \right)^{1/2} = \sqrt{x^* x}, \\
\|x\|_\infty &= \max_{1 \le i \le m} |x_i|, \\
\|x\|_p &= \left(\sum_{i=1}^{m} |x_i|^p \right)^{1/p} \qquad (1 \le p < \infty).
\end{aligned} \tag{3.2}$$



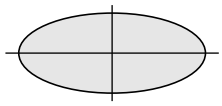
The 2-norm is the Euclidean length function; its unit ball is spherical. The 1-norm is used by airlines to define the maximal allowable size of a suitcase. The Sergel plaza in Stockholm, Sweden has the shape of the unit ball in the 4-norm; the Danish poet Piet Hein popularized this "superellipse" as a pleasing shape for objects such as conference tables.

Aside from the p-norms, the most useful norms are the *weighted p-norms*, where each of the coordinates of a vector space is given its own weight. In general, given any norm $\|\cdot\|$, a weighted norm can be written as

$$\|x\|_W = \|Wx\|. \tag{3.3}$$

Here W is the diagonal matrix in which the ith diagonal entry is the weight $w_i \neq 0$. For example, a weighted 2-norm $\|\cdot\|_W$ on $\mathbb{C}^m$ is specified as follows:

$$\|x\|_W = \left(\sum_{i=1}^{m} |w_i x_i|^2 \right)^{1/2}. \tag{3.4}$$



One can also generalize the idea of weighted norms by allowing W to be an arbitrary nonsingular matrix, not necessarily diagonal (Exercise 3.1).

The most important norms in this book are the unweighted 2-norm and its induced matrix norm.

Matrix Norms Induced by Vector Norms

An $m \times n$ matrix can be viewed as a vector in an mn-dimensional space: each of the mn entries of the matrix is an independent coordinate. Any mn-dimensional norm can therefore be used for measuring the "size" of such a matrix.

However, in dealing with a space of matrices, certain special norms are more useful than the vector norms (3.2)–(3.3) already discussed. These are the *induced matrix norms*, defined in terms of the behavior of a matrix as an operator between its normed domain and range spaces.

Given vector norms $\|\cdot\|_{(n)}$ and $\|\cdot\|_{(m)}$ on the domain and the range of $A \in \mathbb{C}^{m\times n}$, respectively, the induced matrix norm $\|A\|_{(m,n)}$ is the smallest number C for which the following inequality holds for all $x \in \mathbb{C}^n$:

$$\|Ax\|_{(m)} \le C\|x\|_{(n)}. \tag{3.5}$$

In other words, $\|A\|_{(m,n)}$ is the supremum of the ratios $\|Ax\|_{(m)}/\|x\|_{(n)}$ over all vectors $x \in \mathbb{C}^n$—the maximum factor by which A can "stretch" a vector x. We say that $\|\cdot\|_{(m,n)}$ is the matrix norm induced by $\|\cdot\|_{(m)}$ and $\|\cdot\|_{(n)}$.

Because of condition (3) of (3.1), the action of A is determined by its action on unit vectors. Therefore, the matrix norm can be defined equivalently in terms of the images of the unit vectors under A:

$$\|A\|_{(m,n)} = \sup_{\substack{x\in\mathbb{C}^n \\ x\neq 0}} \frac{\|Ax\|_{(m)}}{\|x\|_{(n)}} = \sup_{\substack{x\in\mathbb{C}^n \\ \|x\|_{(n)}=1}} \|Ax\|_{(m)}. \tag{3.6}$$

This form of the definition can be convenient for visualizing induced matrix norms, as in the sketches in (3.2) above.

Examples

Example 3.1. The matrix

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \tag{3.7}$$

maps $\mathbb{C}^2$ to $\mathbb{C}^2$. It also maps $\mathbb{R}^2$ to $\mathbb{R}^2$, which is more convenient if we want to draw pictures and also (it can be shown) sufficient for determining matrix p-norms, since the coefficients of A are real.

Figure 3.1 depicts the action of A on the unit balls of $\mathbb{R}^2$ defined by the 1-, 2-, and ∞-norms. From this figure, one can see a graphical interpretation of these three norms of A. Regardless of the norm, A maps $e_1 = (1,0)^*$ to the first column of A, namely e_1 itself, and $e_2 = (0,1)^*$ to the second column of A, namely $(2,2)^*$. In the 1-norm, the unit vector x that is amplified most by A is $(0,1)^*$ (or its negative), and the amplification factor is 4. In the ∞-norm, the unit vector x that is amplified most by A is $(1,1)^*$ (or its negative), and the amplification factor is 3. In the 2-norm, the unit vector that is amplified most by A is the vector indicated by the dashed line in the figure (or its negative), and the amplification factor is approximately 2.9208. (Note that it must be at least $\sqrt{8} \approx 2.8284$, since $(0,1)^*$ maps to $(2,2)^*$.) We shall consider how to calculate such 2-norm results in Lecture 5. □

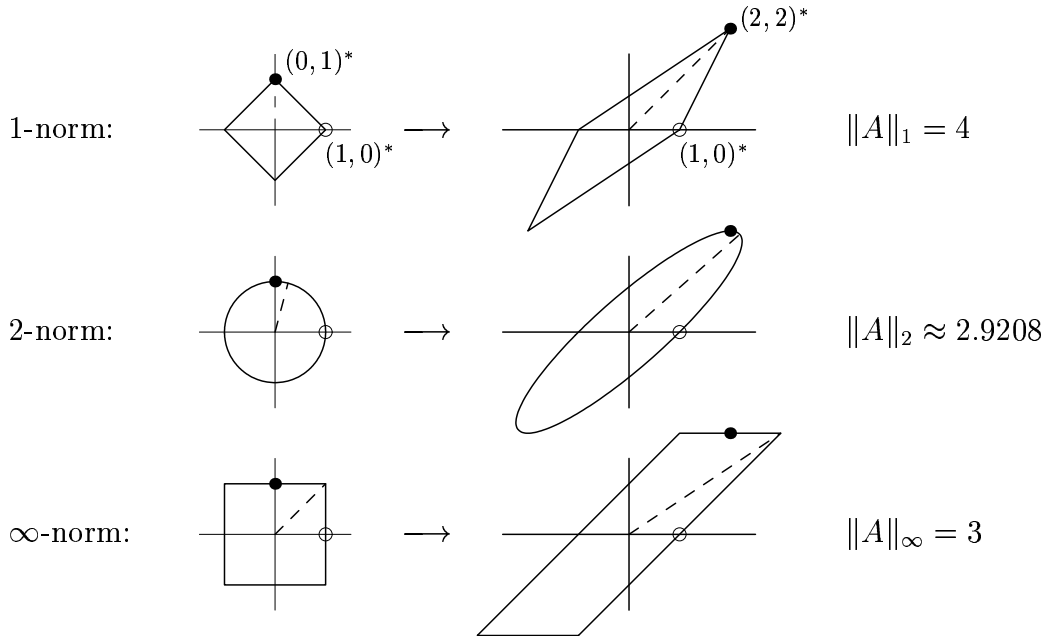


Figure 3.1. *On the left, the unit balls of* $\mathbb{R}^2$ *with respect to* $\|\cdot\|_1$, $\|\cdot\|_2$, *and* $\|\cdot\|_\infty$. *On the right, their images under the matrix* A *of* (3.7). *Dashed lines mark the vectors that are amplified most by* A *in each norm.*

Example 3.2. The *p*-Norm of a Diagonal Matrix. Let D be the diagonal matrix

$$D = \begin{bmatrix} d_1 & & & \\ & d_2 & & \\ & & \ddots & \\ & & & d_m \end{bmatrix}.$$

Then, as in the second row of Figure 3.1, the image of the 2-norm unit sphere under D is an m-dimensional ellipse whose semiaxis lengths are given by the numbers $|d_i|$. The unit vectors amplified most by D are those that are mapped to the longest semiaxis of the ellipse, of length $\max_i\{|d_i|\}$. Therefore, we have $\|D\|_2 = \max_{1\le i\le m}\{|d_i|\}$. In the next lecture we shall see that *every* matrix maps the 2-norm unit sphere to an ellipse—properly called a *hyperellipse* if $m > 2$—though the axes may be oriented arbitrarily.

This result for the 2-norm generalizes to any p: if D is diagonal, then $\|D\|_p = \max_{1\le i\le m} |d_i|$. □

Example 3.3. The 1-Norm of a Matrix. If A is any $m \times n$ matrix, then $\|A\|_1$ is equal to the "maximum column sum" of A. We explain and derive

this result as follows. Write A in terms of its columns

$$A = \left[\begin{array}{c|c|c} & & \\ a_1 & \cdots & a_n \\ & & \end{array}\right], \tag{3.8}$$

where each a_j is an m-vector. Consider the diamond-shaped 1-norm unit ball in $\mathbb{C}^n$, illustrated in (3.2). This is the set $\{x \in \mathbb{C}^n : \sum_{j=1}^n |x_j| \leq 1\}$. Any vector Ax in the image of this set satisfies

$$\|Ax\|_1 \;=\; \Big\|\sum_{j=1}^n x_j a_j\Big\|_1 \;\leq\; \sum_{j=1}^n |x_j|\,\|a_j\|_1 \;\leq\; \max_{1\leq j\leq n} \|a_j\|_1.$$

Therefore, the induced matrix 1-norm satisfies $\|A\|_1 \leq \max_{1\leq j\leq n} \|a_j\|_1$. By choosing $x = e_j$, where j maximizes $\|a_j\|_1$, we attain this bound, and thus the matrix norm is

$$\|A\|_1 \;=\; \max_{1\leq j\leq n} \|a_j\|_1. \tag{3.9}$$

□

Example 3.4. The ∞-Norm of a Matrix. By much the same argument, it can be shown that the ∞-norm of an $m \times n$ matrix is equal to the "maximum row sum,"

$$\|A\|_\infty \;=\; \max_{1\leq i\leq m} \|a_i^*\|_1, \tag{3.10}$$

where a_i^* denotes the ith row of A. □

Cauchy–Schwarz and Hölder Inequalities

Computing matrix p-norms with $p \neq 1, \infty$ is more difficult, and to approach this problem, we note that inner products can be bounded using p-norms. Let p and q satisfy $1/p + 1/q = 1$, with $1 \leq p, q \leq \infty$. Then the *Hölder inequality* states that, for any vectors x and y,

$$|x^* y| \leq \|x\|_p \|y\|_q. \tag{3.11}$$

The *Cauchy–Schwarz inequality* is the special case $p = q = 2$:

$$|x^* y| \leq \|x\|_2 \|y\|_2. \tag{3.12}$$

Derivations of these results can be found in linear algebra texts. Both bounds are tight in the sense that for certain choices of x and y, the inequalities become equalities.

Example 3.5. The 2-Norm of a Row Vector. Consider a matrix A containing a single row. This matrix can be written as $A = a^*$, where a is a column vector. The Cauchy–Schwarz inequality allows us to obtain the induced matrix 2-norm. For any x, we have $\|Ax\|_2 = |a^* x| \leq \|a\|_2 \|x\|_2$. This bound is tight: observe that $\|Aa\|_2 = \|a\|_2^2$. Therefore, we have

$$\|A\|_2 \;=\; \sup_{x\neq 0} \{\|Ax\|_2 / \|x\|_2\} \;=\; \|a\|_2. \qquad \square$$

Example 3.6. The 2-Norm of an Outer Product. More generally, consider the rank-one outer product $A = uv^*$, where u is an m-vector and v is an n-vector. For any n-vector x, we can bound $\|Ax\|_2$ as follows:

$$\|Ax\|_2 = \|uv^*x\|_2 = \|u\|_2|v^*x| \le \|u\|_2\|v\|_2\|x\|_2. \tag{3.13}$$

Therefore $\|A\|_2 \le \|u\|_2\|v\|_2$. Again, this inequality is an equality: consider the case $x = v$. □

Bounding $\|AB\|$ in an Induced Matrix Norm

The induced matrix norm of a matrix product can also be bounded. Let $\|\cdot\|_{(\ell)}$, $\|\cdot\|_{(m)}$, and $\|\cdot\|_{(n)}$ be norms on $\mathbb{C}^l$, $\mathbb{C}^m$, and $\mathbb{C}^n$, respectively, and let A be an $l \times m$ matrix and B an $m \times n$ matrix. For any $x \in \mathbb{C}^n$ we have

$$\|ABx\|_{(\ell)} \le \|A\|_{(\ell,m)}\|Bx\|_{(m)} \le \|A\|_{(\ell,m)}\|B\|_{(m,n)}\|x\|_{(n)}.$$

Therefore the induced norm of AB must satisfy

$$\|AB\|_{(\ell,n)} \le \|A\|_{(\ell,m)}\|B\|_{(m,n)}. \tag{3.14}$$

In general, this inequality is not an equality. For example, the inequality $\|A^n\| \le \|A\|^n$ holds for any square matrix in any matrix norm induced by a vector norm, but $\|A^n\| = \|A\|^n$ does not hold in general for $n \ge 2$.

General Matrix Norms

As noted above, matrix norms do not have to be induced by vector norms. In general, a matrix norm must merely satisfy the three vector norm conditions (3.1) applied in the mn-dimensional vector space of matrices:

$$\begin{aligned} &(1)\ \ \|A\| \ge 0, \text{ and } \|A\| = 0 \text{ only if } A = 0, \\ &(2)\ \ \|A + B\| \le \|A\| + \|B\|, \\ &(3)\ \ \|\alpha A\| = |\alpha|\,\|A\|. \end{aligned} \tag{3.15}$$

The most important matrix norm which is not induced by a vector norm is the *Hilbert–Schmidt* or *Frobenius norm*, defined by

$$\|A\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2}. \tag{3.16}$$

Observe that this is the same as the 2-norm of the matrix when viewed as an mn-dimensional vector. The formula for the Frobenius norm can also be

written in terms of individual rows or columns. For example, if a_j is the jth column of A, we have

$$\|A\|_F = \left(\sum_{j=1}^{n} \|a_j\|_2^{\,2} \right)^{1/2}. \tag{3.17}$$

This identity, as well as the analogous result based on rows instead of columns, can be expressed compactly by the equation

$$\|A\|_F = \sqrt{\mathrm{tr}(A^*A)} = \sqrt{\mathrm{tr}(AA^*)}, \tag{3.18}$$

where $\mathrm{tr}(B)$ denotes the *trace* of B, the sum of its diagonal entries.

Like an induced matrix norm, the Frobenius norm can be used to bound products of matrices. Let $C = AB$ with entries c_{ik}, and let a_i^* denote the ith row of A and b_j the jth column of B. Then $c_{ij} = a_i^* b_j$, so by the Cauchy–Schwarz inequality we have $|c_{ij}| \le \|a_i\|_2 \, \|b_j\|_2$. Squaring both sides and summing over all i, j, we obtain

$$\begin{aligned} \|AB\|_F^2 &= \sum_{i=1}^{n} \sum_{j=1}^{m} |c_{ij}|^2 \\ &\le \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\|a_i\|_2 \, \|b_j\|_2 \right)^2 \\ &= \sum_{i=1}^{n} \left(\|a_i\|_2 \right)^2 \sum_{j=1}^{m} \left(\|b_j\|_2 \right)^2 \quad = \quad \|A\|_F^2 \, \|B\|_F^2. \end{aligned}$$

Invariance under Unitary Multiplication

One of the many special properties of the matrix 2-norm is that, like the vector 2-norm, it is invariant under multiplication by unitary matrices. The same property holds for the Frobenius norm.

Theorem 3.1. *For any $A \in \mathbb{C}^{m \times n}$ and unitary $Q \in \mathbb{C}^{m \times m}$, we have*

$$\|QA\|_2 = \|A\|_2, \qquad \|QA\|_F = \|A\|_F.$$

Proof. Since $\|Qx\|_2 = \|x\|_2$ for every x, by (2.10), the invariance in the 2-norm follows from (3.6). For the Frobenius norm we may use (3.18). □

Theorem 3.1 remains valid if Q is generalized to a rectangular matrix with orthonormal columns, that is, $Q \in \mathbb{C}^{p \times m}$ with $p > m$. Analogous identities also hold for multiplication by unitary matrices on the right, or more generally, by rectangular matrices with orthonormal rows.

Exercises

3.1. Prove that if W is an arbitrary nonsingular matrix, the function $\|\cdot\|_W$ defined by (3.3) is a vector norm.

3.2. Let $\|\cdot\|$ denote any norm on $\mathbb{C}^m$ and also the induced matrix norm on $\mathbb{C}^{m\times m}$. Show that $\rho(A) \le \|A\|$, where $\rho(A)$ is the *spectral radius* of A, i.e., the largest absolute value $|\lambda|$ of an eigenvalue λ of A.

3.3. Vector and matrix p-norms are related by various inequalities, often involving the dimensions m or n. For each of the following, verify the inequality and give an example of a nonzero vector or matrix (for general m, n) for which equality is achieved. In this problem x is an m-vector and A is an $m \times n$ matrix.

(a) $\|x\|_\infty \le \|x\|_2$,

(b) $\|x\|_2 \le \sqrt{m}\,\|x\|_\infty$,

(c) $\|A\|_\infty \le \sqrt{n}\,\|A\|_2$,

(d) $\|A\|_2 \le \sqrt{m}\,\|A\|_\infty$.

3.4. Let A be an $m \times n$ matrix and let B be a submatrix of A, that is, a $\mu \times \nu$ matrix ($\mu \le m$, $\nu \le n$) obtained by selecting certain rows and columns of A.

(a) Explain how B can be obtained by multiplying A by certain row and column "deletion matrices" as in step 7 of Exercise 1.1.

(b) Using this product, show that $\|B\|_p \le \|A\|_p$ for any p with $1 \le p \le \infty$.

3.5. Example 3.6 shows that if E is an outer product $E = uv^*$, then $\|E\|_2 = \|u\|_2\|v\|_2$. Is the same true for the Frobenius norm, i.e., $\|E\|_F = \|u\|_F\|v\|_F$? Prove it or give a counterexample.

3.6. Let $\|\cdot\|$ denote any norm on $\mathbb{C}^m$. The corresponding *dual norm* $\|\cdot\|'$ is defined by the formula $\|x\|' = \sup_{\|y\|=1} |y^*x|$.

(a) Prove that $\|\cdot\|'$ is a norm.

(b) Let $x, y \in \mathbb{C}^m$ with $\|x\| = \|y\| = 1$ be given. Show that there exists a rank-one matrix $B = yz^*$ such that $Bx = y$ and $\|B\| = 1$, where $\|B\|$ is the matrix norm of B induced by the vector norm $\|\cdot\|$. You may use the following lemma, without proof: given $x \in \mathbb{C}^m$, there exists a nonzero $z \in \mathbb{C}^m$ such that $|z^*x| = \|z\|'\|x\|$.

Lecture 4. The Singular Value Decomposition

The singular value decomposition (SVD) is a matrix factorization whose computation is a step in many algorithms. Equally important is the use of the SVD for conceptual purposes. Many problems of linear algebra can be better understood if we first ask the question: what if we take the SVD?

A Geometric Observation

The SVD is motivated by the following geometric fact:

> The image of the unit sphere under any $m \times n$ matrix is a hyperellipse.

The SVD is applicable to both real and complex matrices. However, in describing the geometric interpretation, we assume as usual that the matrix is real.

The term "hyperellipse" may be unfamiliar, but this is just the m-dimensional generalization of an ellipse. We may define a hyperellipse in $\mathbb{R}^m$ as the surface obtained by stretching the unit sphere in $\mathbb{R}^m$ by some factors $\sigma_1, \ldots, \sigma_m$ (possibly zero) in some orthogonal directions $u_1, \ldots, u_m \in \mathbb{R}^m$. For convenience, let us take the u_i to be unit vectors, i.e., $\|u_i\|_2 = 1$. The vectors $\{\sigma_i u_i\}$ are the *principal semiaxes* of the hyperellipse, with lengths $\sigma_1, \ldots, \sigma_m$. If A has rank r, exactly r of the lengths σ_i will turn out to be nonzero, and in particular, if $m \geq n$, at most n of them will be nonzero.

Our opening statement about the image of the unit sphere has the following meaning. By the unit sphere, we mean the usual Euclidean sphere in n-space, i.e., the unit sphere in the 2-norm; let us denote it by S. Then AS, the image of S under the mapping A, is a hyperellipse as just defined.

This geometric fact is not obvious. We shall restate it in the language of linear algebra and prove it later. For the moment, assume it is true.

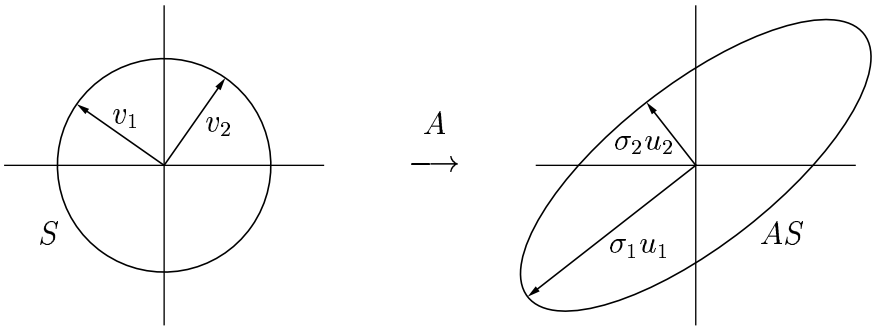


Figure 4.1. *SVD of a* 2×2 *matrix.*

Let S be the unit sphere in $\mathbb{R}^n$, and take any $A \in \mathbb{R}^{m \times n}$ with $m \geq n$. For simplicity, suppose for the moment that A has full rank n. The image AS is a hyperellipse in $\mathbb{R}^m$. We now define some properties of A in terms of the shape of AS. The key ideas are indicated in Figure 4.1.

First, we define the n *singular values* of A. These are the lengths of the n principal semiaxes of AS, written $\sigma_1, \sigma_2, \ldots, \sigma_n$. It is conventional to assume that the singular values are numbered in descending order, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n > 0$.

Next, we define the n *left singular vectors* of A. These are the unit vectors $\{u_1, u_2, \ldots, u_n\}$ oriented in the directions of the principal semiaxes of AS, numbered to correspond with the singular values. Thus the vector $\sigma_i u_i$ is the ith largest principal semiaxis of AS.

Finally, we define the n *right singular vectors* of A. These are the unit vectors $\{v_1, v_2, \ldots, v_n\} \in S$ that are the preimages of the principal semiaxes of AS, numbered so that $Av_j = \sigma_j u_j$.

The terms "left" and "right" in the definitions above are decidedly awkward. They come from the positions of the factors U and V in (4.2) and (4.3), below. What is awkward is that in a sketch like Figure 4.1, the left singular vectors correspond to the space on the right, and the right singular vectors correspond to the space on the left! One could resolve this problem by interchanging the two halves of the figure, with the map A pointing from right to left, but that would go against deeply ingrained habits.

Reduced SVD

We have just mentioned that the equations relating right singular vectors $\{v_j\}$ and left singular vectors $\{u_j\}$ can be written

$$Av_j = \sigma_j u_j, \qquad 1 \le j \le n. \tag{4.1}$$

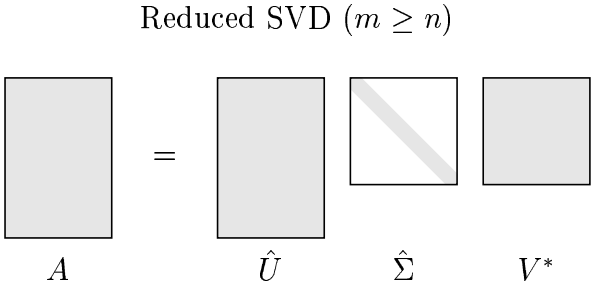
This collection of vector equations can be expressed as a matrix equation,

$$\begin{bmatrix} & & \\ & A & \\ & & \end{bmatrix} \begin{bmatrix} & & & \\ v_1 & v_2 & \cdots & v_n \\ & & & \end{bmatrix} = \begin{bmatrix} & & & \\ u_1 & u_2 & \cdots & u_n \\ & & & \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_n \end{bmatrix},$$

or, more compactly, $AV = \hat{U}\hat{\Sigma}$. In this matrix equation, $\hat{\Sigma}$ is an $n \times n$ diagonal matrix with positive real entries (since A was assumed to have full rank n), $\hat{U}$ is an $m \times n$ matrix with orthonormal columns, and V is an $n \times n$ matrix with orthonormal columns. Thus V is unitary, and we can multiply on the right by its inverse V^* to obtain

$$A = \hat{U}\hat{\Sigma}V^*. \tag{4.2}$$

This factorization of A is called a *reduced singular value decomposition*, or *reduced SVD*, of A. Schematically, it looks like this:



Full SVD

In most applications, the SVD is used in exactly the form just described. However, this is not the way in which the idea of an SVD is usually formulated in textbooks. We have introduced the term "reduced" and the hats on U and Σ in order to distinguish the factorization (4.2) from the more standard "full" SVD. This "reduced" vs. "full" terminology and hatted notation will be maintained throughout the book, and we shall make a similar distinction between reduced and full QR factorizations. Reminders of these conventions are printed on the inside front cover.

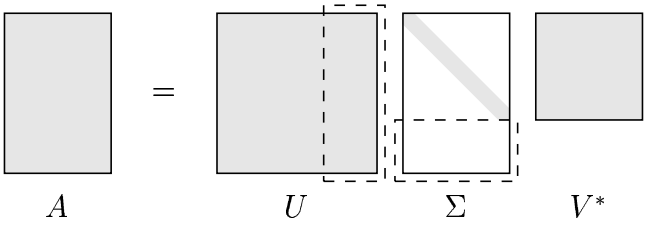
The idea is as follows. The columns of $\hat{U}$ are n orthonormal vectors in the m-dimensional space $\mathbb{C}^m$. Unless $m = n$, they do not form a basis of $\mathbb{C}^m$, nor is $\hat{U}$ a unitary matrix. However, by adjoining an additional $m - n$ orthonormal columns, $\hat{U}$ can be extended to a unitary matrix. Let us do this in an arbitrary fashion, and call the result U.

If $\hat{U}$ is replaced by U in (4.2), then $\hat{\Sigma}$ will have to change too. For the product to remain unaltered, the last $m-n$ columns of U should be multiplied by zero. Accordingly, let Σ be the $m \times n$ matrix consisting of $\hat{\Sigma}$ in the upper $n \times n$ block together with $m - n$ rows of zeros below. We now have a new factorization, the *full SVD* of A:

$$A = U\Sigma V^*. \tag{4.3}$$

Here U is $m \times m$ and unitary, V is $n \times n$ and unitary, and Σ is $m \times n$ and diagonal with positive real entries. Schematically:

Full SVD ($m \geq n$)



The dashed lines indicate the "silent" columns of U and rows of Σ that are discarded in passing from (4.3) to (4.2).

Having described the full SVD, we can now discard the simplifying assumption that A has full rank. If A is rank-deficient, the factorization (4.3) is still appropriate. All that changes is that now not n but only r of the left singular vectors of A are determined by the geometry of the hyperellipse. To construct the unitary matrix U, we introduce $m - r$ instead of just $m - n$ additional arbitrary orthonormal columns. The matrix V will also need $n - r$ arbitrary orthonormal columns to extend the r columns determined by the geometry. The matrix Σ will now have r positive diagonal entries, with the remaining $n - r$ equal to zero.

By the same token, the reduced SVD (4.2) also makes sense for matrices A of less than full rank. One can take $\hat{U}$ to be $m \times n$, with $\hat{\Sigma}$ of dimensions $n \times n$ with some zeros on the diagonal, or further compress the representation so that $\hat{U}$ is $m \times r$ and $\hat{\Sigma}$ is $r \times r$ and strictly positive on the diagonal.

Formal Definition

Let m and n be arbitrary; we do not require $m \geq n$. Given $A \in \mathbb{C}^{m \times n}$, not necessarily of full rank, a *singular value decomposition* (SVD) of A is a

factorization

$$A = U\Sigma V^* \tag{4.4}$$

where

$$\begin{aligned} U &\in \mathbb{C}^{m\times m} \quad \text{is unitary,} \\ V &\in \mathbb{C}^{n\times n} \quad \text{is unitary,} \\ \Sigma &\in \mathbb{R}^{m\times n} \quad \text{is diagonal.} \end{aligned}$$

In addition, it is assumed that the diagonal entries σ_j of Σ are nonnegative and in nonincreasing order; that is, $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_p \geq 0$, where $p = \min(m, n)$.

Note that the diagonal matrix Σ has the same shape as A even when A is not square, but U and V are always square unitary matrices.

It is clear that the image of the unit sphere in $\mathbb{R}^n$ under a map $A = U\Sigma V^*$ must be a hyperellipse in $\mathbb{R}^m$. The unitary map V^* preserves the sphere, the diagonal matrix Σ stretches the sphere into a hyperellipse aligned with the canonical basis, and the final unitary map U rotates or reflects the hyperellipse without changing its shape. Thus, if we can prove that every matrix has an SVD, we shall have proved that the image of the unit sphere under any linear map is a hyperellipse, as claimed at the outset of this lecture.

Existence and Uniqueness

Theorem 4.1. *Every matrix $A \in \mathbb{C}^{m\times n}$ has a singular value decomposition* (4.4)*. Furthermore, the singular values $\{\sigma_j\}$ are uniquely determined, and, if A is square and the σ_j are distinct, the left and right singular vectors $\{u_j\}$ and $\{v_j\}$ are uniquely determined up to complex signs (i.e., complex scalar factors of absolute value* 1*).*

Proof. To prove existence of the SVD, we isolate the direction of the largest action of A, and then proceed by induction on the dimension of A.

Set $\sigma_1 = \|A\|_2$. By a compactness argument, there must be vectors $v_1 \in \mathbb{C}^n$ and $u_1 \in \mathbb{C}^m$ with $\|v_1\|_2 = \|u_1\|_2 = 1$ and $Av_1 = \sigma_1 u_1$. Consider any extensions of v_1 to an orthonormal basis $\{v_j\}$ of $\mathbb{C}^n$ and of u_1 to an orthonormal basis $\{u_j\}$ of $\mathbb{C}^m$, and let U_1 and V_1 denote the unitary matrices with columns u_j and v_j, respectively. Then we have

$$U_1^* A V_1 = S = \begin{bmatrix} \sigma_1 & w^* \\ 0 & B \end{bmatrix}, \tag{4.5}$$

where 0 is a column vector of dimension $m-1$, w^* is a row vector of dimension $n-1$, and B has dimensions $(m-1)\times(n-1)$. Furthermore,

$$\left\| \begin{bmatrix} \sigma_1 & w^* \\ 0 & B \end{bmatrix} \begin{bmatrix} \sigma_1 \\ w \end{bmatrix} \right\|_2 \geq \sigma_1^2 + w^* w = (\sigma_1^2 + w^* w)^{1/2} \left\| \begin{bmatrix} \sigma_1 \\ w \end{bmatrix} \right\|_2,$$

implying $\|S\|_2 \geq (\sigma_1^2 + w^*w)^{1/2}$. Since U_1 and V_1 are unitary, we know that $\|S\|_2 = \|A\|_2 = \sigma_1$, so this implies $w = 0$.

If $n = 1$ or $m = 1$, we are done. Otherwise, the submatrix B describes the action of A on the subspace orthogonal to v_1. By the induction hypothesis, B has an SVD $B = U_2\Sigma_2 V_2^*$. Now it is easily verified that

$$A = U_1 \begin{bmatrix} 1 & 0 \\ 0 & U_2 \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 \\ 0 & \Sigma_2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & V_2 \end{bmatrix}^* V_1^*$$

is an SVD of A, completing the proof of existence.

For the uniqueness claim, the geometric justification is straightforward: if the semiaxis lengths of a hyperellipse are distinct, then the semiaxes themselves are determined by the geometry, up to signs. Algebraically, we can argue as follows. First we note that σ_1 is uniquely determined by the condition that it is equal to $\|A\|_2$, as follows from (4.4). Now suppose that in addition to v_1, there is another linearly independent vector w with $\|w\|_2 = 1$ and $\|Aw\|_2 = \sigma_1$. Define a unit vector v_2, orthogonal to v_1, as a linear combination of v_1 and w,

$$v_2 = \frac{w - (v_1^*w)v_1}{\|w - (v_1^*w)v_1\|_2}.$$

Since $\|A\|_2 = \sigma_1$, $\|Av_2\|_2 \leq \sigma_1$; but this must be an equality, for otherwise, since $w = v_1c + v_2s$ for some constants c and s with $|c|^2 + |s|^2 = 1$, we would have $\|Aw\|_2 < \sigma_1$. This vector v_2 is a second right singular vector of A corresponding to the singular value σ_1; it will lead to the appearance of a vector y (equal to the last $n-1$ components of $V_1^*v_2$) with $\|y\|_2 = 1$ and $\|By\|_2 = \sigma_1$. We conclude that, if the singular vector v_1 is not unique, then the corresponding singular value σ_1 is not simple. To complete the uniqueness proof we note that, as indicated above, once σ_1, v_1, and u_1 are determined, the remainder of the SVD is determined by the action of A on the space orthogonal to v_1. Since v_1 is unique up to sign, this orthogonal space is uniquely defined, and the uniqueness of the remaining singular values and vectors now follows by induction. □

Exercises

4.1. Determine SVDs of the following matrices (by hand calculation):

(a) $\begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$, (b) $\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, (c) $\begin{bmatrix} 0 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$, (d) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$, (e) $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

4.2. Suppose A is an $m \times n$ matrix and B is the $n \times m$ matrix obtained by rotating A ninety degrees clockwise on paper (not exactly a standard mathematical transformation!). Do A and B have the same singular values? Prove that the answer is yes or give a counterexample.

4.3. Write a MATLAB program (see Lecture 9) which, given a real 2×2 matrix A, plots the right singular vectors v_1 and v_2 in the unit circle and also the left singular vectors u_1 and u_2 in the appropriate ellipse, as in Figure 4.1. Apply your program to the matrix (3.7) and also to the 2×2 matrices of Exercise 4.1.

4.4. Two matrices $A, B \in \mathbb{C}^{m \times m}$ are *unitarily equivalent* if $A = QBQ^*$ for some unitary $Q \in \mathbb{C}^{m \times m}$. Is it true or false that A and B are unitarily equivalent if and only if they have the same singular values?

4.5. Theorem 4.1 asserts that every $A \in \mathbb{C}^{m \times n}$ has an SVD $A = U\Sigma V^*$. Show that if A is real, then it has a real SVD ($U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$).

Lecture 5. More on the SVD

We continue our discussion of the singular value decomposition, emphasizing its connection with low-rank approximation of matrices in the 2-norm and the Frobenius norm.

A Change of Bases

The SVD makes it possible for us to say that every matrix is diagonal—if only one uses the proper bases for the domain and range spaces.

Here is how the change of bases works. Any $b \in \mathbb{C}^m$ can be expanded in the basis of left singular vectors of A (columns of U), and any $x \in \mathbb{C}^n$ can be expanded in the basis of right singular vectors of A (columns of V). The coordinate vectors for these expansions are

$$b' = U^* b, \qquad x' = V^* x.$$

By (4.3), the relation $b = Ax$ can be expressed in terms of b' and x':

$$b = Ax \iff U^* b = U^* A x = U^* U \Sigma V^* x \iff b' = \Sigma x'.$$

Whenever $b = Ax$, we have $b' = \Sigma x'$. Thus A reduces to the diagonal matrix Σ when the range is expressed in the basis of columns of U and the domain is expressed in the basis of columns of V.

SVD vs. Eigenvalue Decomposition

The theme of diagonalizing a matrix by expressing it in terms of a new basis also underlies the study of eigenvalues. A nondefective square matrix A can be expressed as a diagonal matrix of eigenvalues Λ, if the range and domain are represented in a basis of eigenvectors.

If the columns of a matrix $X \in \mathbb{C}^{m\times m}$ contain linearly independent eigenvectors of $A \in \mathbb{C}^{m\times m}$, the *eigenvalue decomposition* of A is

$$A = X\Lambda X^{-1}, \tag{5.1}$$

where Λ is an $m \times m$ diagonal matrix whose entries are the eigenvalues of A. This implies that if we define, for $b, x \in \mathbb{C}^m$ satisfying $b = Ax$,

$$b' = X^{-1}b, \qquad x' = X^{-1}x,$$

then the newly expanded vectors b' and x' satisfy $b' = \Lambda x'$. Eigenvalues are treated systematically in Lecture 24.

There are fundamental differences between the SVD and the eigenvalue decomposition. One is that the SVD uses two different bases (the sets of left and right singular vectors), whereas the eigenvalue decomposition uses just one (the eigenvectors). Another is that the SVD uses orthonormal bases, whereas the eigenvalue decomposition uses a basis that generally is not orthogonal. A third is that not all matrices (even square ones) have an eigenvalue decomposition, but all matrices (even rectangular ones) have a singular value decomposition, as we established in Theorem 4.1. In applications, eigenvalues tend to be relevant to problems involving the behavior of iterated forms of A, such as matrix powers A^k or exponentials e^{tA}, whereas singular vectors tend to be relevant to problems involving the behavior of A itself, or its inverse.

Matrix Properties via the SVD

The power of the SVD becomes apparent as we begin to catalogue its connections with other fundamental topics of linear algebra. For the following theorems, assume that A has dimensions $m \times n$. Let p be the minimum of m and n, let $r \le p$ denote the number of nonzero singular values of A, and let $\langle x, y, \ldots, z\rangle$ denote the space spanned by the vectors $x, y, \ldots, z$.

Theorem 5.1. *The rank of A is r, the number of nonzero singular values.*

Proof. The rank of a diagonal matrix is equal to the number of its nonzero entries, and in the decomposition $A = U\Sigma V^*$, U and V are of full rank. Therefore $\operatorname{rank}(A) = \operatorname{rank}(\Sigma) = r$. □

Theorem 5.2. $\operatorname{range}(A) = \langle u_1, \ldots, u_r\rangle$ *and* $\operatorname{null}(A) = \langle v_{r+1}, \ldots, v_n\rangle$.

Proof. This is a consequence of the fact that $\operatorname{range}(\Sigma) = \langle e_1, \ldots, e_r\rangle \subseteq \mathbb{C}^m$ and $\operatorname{null}(\Sigma) = \langle e_{r+1}, \ldots, e_n\rangle \subseteq \mathbb{C}^n$. □

Theorem 5.3. *$\|A\|_2 = \sigma_1$ and $\|A\|_F = \sqrt{\sigma_1^2 + \sigma_2^2 + \cdots + \sigma_r^2}$.*

Proof. The first result was already established in the proof of Theorem 4.1: since $A = U\Sigma V^*$ with unitary U and V, $\|A\|_2 = \|\Sigma\|_2 = \max\{|\sigma_j|\} = \sigma_1$, by Theorem 3.1. For the second, note that by Theorem 3.1 and the remark following, the Frobenius norm is invariant under unitary multiplication, so $\|A\|_F = \|\Sigma\|_F$, and by (3.16), this is given by the stated formula. □

Theorem 5.4. *The nonzero singular values of A are the square roots of the nonzero eigenvalues of A^*A or AA^*. (These matrices have the same nonzero eigenvalues.)*

Proof. From the calculation

$$A^*A = (U\Sigma V^*)^*(U\Sigma V^*) = V\Sigma^* U^* U\Sigma V^* = V(\Sigma^*\Sigma)V^*,$$

we see that A^*A is similar to $\Sigma^*\Sigma$ and hence has the same n eigenvalues (see Lecture 24). The eigenvalues of the diagonal matrix $\Sigma^*\Sigma$ are $\sigma_1^2, \sigma_2^2, \ldots, \sigma_p^2$, with $n - p$ additional zero eigenvalues if $n > p$. A similar calculation applies to the m eigenvalues of AA^*. □

Theorem 5.5. *If $A = A^*$, then the singular values of A are the absolute values of the eigenvalues of A.*

Proof. As is well known (see Exercise 2.3), a hermitian matrix has a complete set of orthogonal eigenvectors, and all of the eigenvalues are real. An equivalent statement is that (5.1) holds with X equal to some unitary matrix Q and Λ a real diagonal matrix. But then we can write

$$A = Q\Lambda Q^* = Q|\Lambda|\,\mathrm{sign}(\Lambda)\,Q^*, \tag{5.2}$$

where $|\Lambda|$ and $\mathrm{sign}(\Lambda)$ denote the diagonal matrices whose entries are the numbers $|\lambda_j|$ and $\mathrm{sign}(\lambda_j)$, respectively. (We could equally well have put the factor $\mathrm{sign}(\Lambda)$ on the left of $|\Lambda|$ instead of the right.) Since $\mathrm{sign}(\Lambda)Q^*$ is unitary whenever Q is unitary, (5.2) is an SVD of A, with the singular values equal to the diagonal entries of $|\Lambda|$, $|\lambda_j|$. If desired, these numbers can be put into nonincreasing order by inserting suitable permutation matrices as factors in the left-hand unitary matrix of (5.2), Q, and the right-hand unitary matrix, $\mathrm{sign}(\Lambda)Q^*$. □

Theorem 5.6. *For $A \in \mathbb{C}^{m\times m}$, $|\det(A)| = \prod_{i=1}^{m} \sigma_i$.*

Proof. The determinant of a product of square matrices is the product of the determinants of the factors. Furthermore, the determinant of a unitary matrix is always 1 in absolute value; this follows from the formula $U^*U = I$ and the property $\det(U^*) = (\det(U))^*$. Therefore,

$$|\det(A)| = |\det(U\Sigma V^*)| = |\det(U)|\,|\det(\Sigma)|\,|\det(V^*)| = |\det(\Sigma)| = \prod_{i=1}^{m} \sigma_i.$$

□

Low-Rank Approximations

But what *is* the SVD? Another approach to an explanation is to consider how a matrix A might be represented as a sum of rank-one matrices.

Theorem 5.7. *A is the sum of r rank-one matrices:*

$$A = \sum_{j=1}^{r} \sigma_j u_j v_j^*. \tag{5.3}$$

Proof. If we write Σ as a sum of r matrices Σ_j, where $\Sigma_j = \mathrm{diag}(0, \ldots, 0, \sigma_j, 0, \ldots, 0)$, then (5.3) follows from (4.3). □

There are many ways to express an $m \times n$ matrix A as a sum of rank-one matrices. For example, A could be written as the sum of its m rows, or its n columns, or its mn entries. For another example, Gaussian elimination reduces A to the sum of a full rank-one matrix, a rank-one matrix whose first row and column are zero, a rank-one matrix whose first two rows and columns are zero, and so on.

Formula (5.3), however, represents a decomposition into rank-one matrices with a deeper property: *the νth partial sum captures as much of the energy of A as possible*. This statement holds with "energy" defined by either the 2-norm or the Frobenius norm. We can make it precise by formulating a problem of best approximation of a matrix A by matrices of lower rank.

Theorem 5.8. *For any ν with $0 \le \nu \le r$, define*

$$A_\nu = \sum_{j=1}^{\nu} \sigma_j u_j v_j^*; \tag{5.4}$$

if $\nu = p = \min\{m, n\}$, define $\sigma_{\nu+1} = 0$. Then

$$\|A - A_\nu\|_2 \;=\; \inf_{\substack{B \in \mathbb{C}^{m\times n} \\ \mathrm{rank}(B) \le \nu}} \|A - B\|_2 \;=\; \sigma_{\nu+1}.$$

Proof. Suppose there is some B with $\text{rank}(B) \le \nu$ such that $\|A - B\|_2 < \|A - A_\nu\|_2 = \sigma_{\nu+1}$. Then there is an $(n-\nu)$-dimensional subspace $W \subseteq \mathbb{C}^n$ such that $w \in W \Rightarrow Bw = 0$. Accordingly, for any $w \in W$, we have $Aw = (A-B)w$ and

$$\|Aw\|_2 = \|(A-B)w\|_2 \le \|A-B\|_2 \, \|w\|_2 < \sigma_{\nu+1}\|w\|_2.$$

Thus W is an $(n-\nu)$-dimensional subspace where $\|Aw\| < \sigma_{\nu+1}\|w\|$. But there is a $(\nu+1)$-dimensional subspace where $\|Aw\| \ge \sigma_{\nu+1}\|w\|$, namely the space spanned by the first $\nu+1$ right singular vectors of A. Since the sum of the dimensions of these spaces exceeds n, there must be a nonzero vector lying in both, and this is a contradiction. □

Theorem 5.8 has a geometric interpretation. What is the best approximation of a hyperellipsoid by a line segment? Take the line segment to be the longest axis. What is the best approximation by a two-dimensional ellipsoid? Take the ellipsoid spanned by the longest and the second-longest axis. Continuing in this fashion, at each step we improve the approximation by adding into our approximation the largest axis of the hyperellipsoid not yet included. After r steps, we have captured all of A. This idea has ramifications in areas as disparate as image compression (see Exercise 9.3) and functional analysis.

We state the analogous result for the Frobenius norm without proof.

Theorem 5.9. *For any ν with $0 \le \nu \le r$, the matrix A_ν of (5.4) also satisfies*

$$\|A - A_\nu\|_F \;=\; \inf_{\substack{B \in \mathbb{C}^{m\times n} \\ \text{rank}(B) \le \nu}} \|A - B\|_F \;=\; \sqrt{\sigma_{\nu+1}^2 + \cdots + \sigma_r^2}.$$

Computation of the SVD

In this and the previous lecture, we have examined the properties of the SVD but not considered how it can be computed. As it happens, the computation of the SVD is a fascinating subject. The best methods are variants of algorithms used for computing eigenvalues, and we shall discuss them in Lecture 31.

Once one can compute it, the SVD can be used as a tool for all kinds of problems. In fact, most of the theorems of this lecture have computational consequences. The best method for determining the rank of a matrix is to count the number of singular values greater than a judiciously chosen tolerance (Theorem 5.1). The most accurate method for finding an orthonormal basis of a range or a nullspace is via Theorem 5.2. (For both of these examples, QR factorization provides alternative algorithms that are faster but not always as accurate.) Theorem 5.3 represents the standard method for computing $\|A\|_2$, and Theorems 5.8 and 5.9, the standards for computing low-rank approximations with respect to $\|\cdot\|_2$ and $\|\cdot\|_F$. Besides these examples, the SVD is also an ingredient in robust algorithms for least squares fitting, intersection of subspaces, regularization, and numerous other problems.

Exercises

5.1. In Example 3.1 we considered the matrix (3.7) and asserted, among other things, that its 2-norm is approximately 2.9208. Using the SVD, work out (on paper) the exact values of $\sigma_{\min}(A)$ and $\sigma_{\max}(A)$ for this matrix.

5.2. Using the SVD, prove that any matrix in $\mathbb{C}^{m\times n}$ is the limit of a sequence of matrices of full rank. In other words, prove that the set of full-rank matrices is a dense subset of $\mathbb{C}^{m\times n}$. Use the 2-norm for your proof. (The norm doesn't matter, since all norms on a finite-dimensional space are equivalent.)

5.3. Consider the matrix

$$A = \begin{bmatrix} -2 & 11 \\ -10 & 5 \end{bmatrix}.$$

(a) Determine, on paper, a real SVD of A in the form $A = U\Sigma V^T$. The SVD is not unique, so find the one that has the minimal number of minus signs in U and V.

(b) List the singular values, left singular vectors, and right singular vectors of A. Draw a careful, labeled picture of the unit ball in $\mathbb{R}^2$ and its image under A, together with the singular vectors, with the coordinates of their vertices marked.

(c) What are the 1-, 2-, ∞-, and Frobenius norms of A?

(d) Find A^{-1} not directly, but via the SVD.

(e) Find the eigenvalues λ_1, λ_2 of A.

(f) Verify that $\det A = \lambda_1\lambda_2$ and $|\det A| = \sigma_1\sigma_2$.

(g) What is the area of the ellipsoid onto which A maps the unit ball of $\mathbb{R}^2$?

5.4. Suppose $A \in \mathbb{C}^{m\times m}$ has an SVD $A = U\Sigma V^*$. Find an eigenvalue decomposition (5.1) of the $2m \times 2m$ hermitian matrix

$$\begin{bmatrix} 0 & A^* \\ A & 0 \end{bmatrix}.$$